Lab 5: Static Code Analysis

Pylint, Bandit, and Flake8 — Output Screenshots

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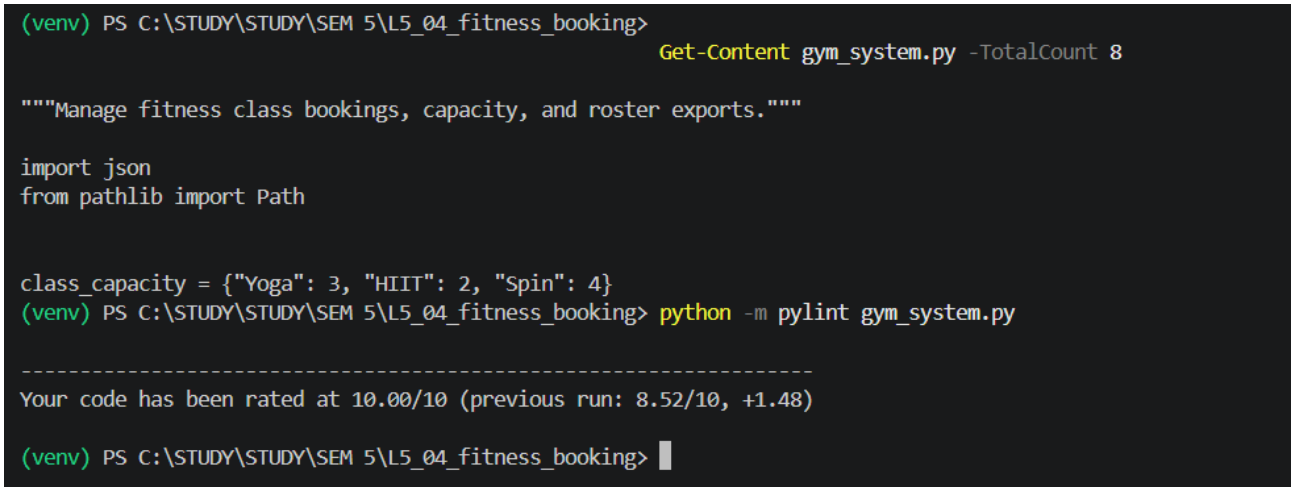
SECTION: G

Final results

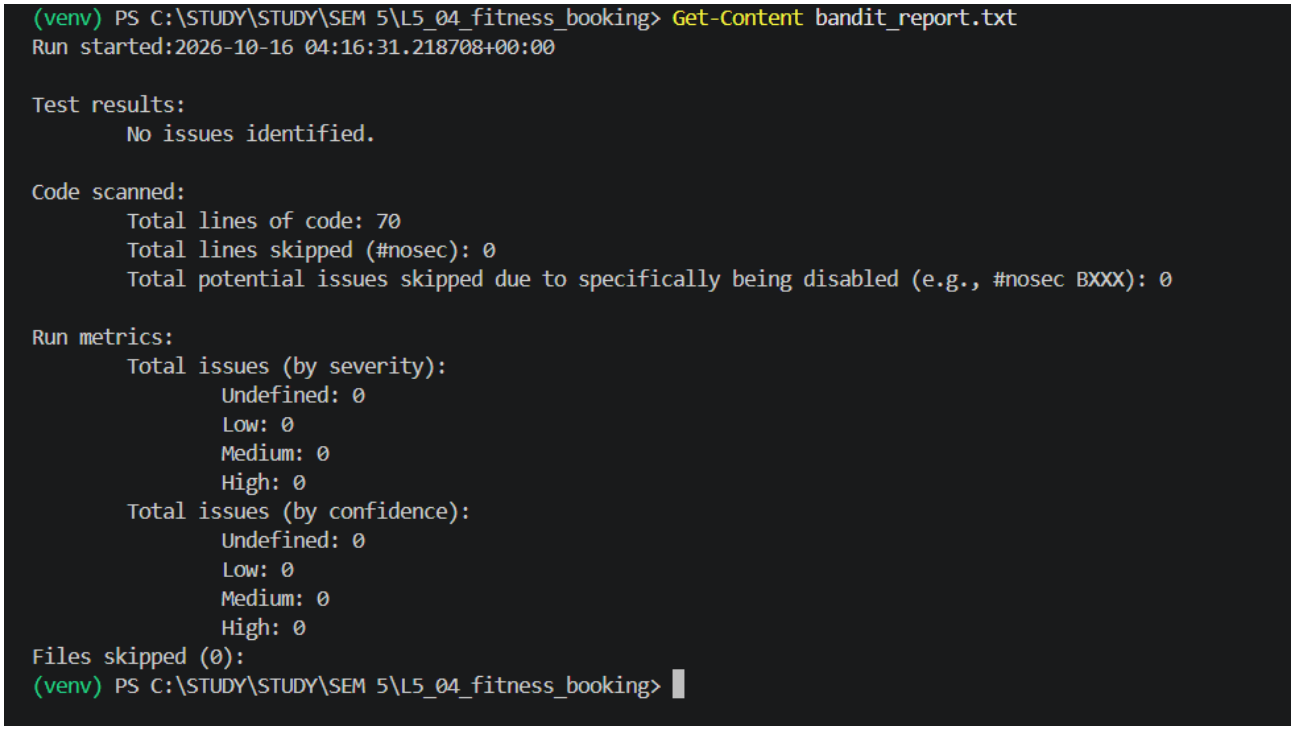
- Pylint: 10.00/10 (as shown in the final terminal screenshot).
- Bandit: No issues identified in the final report.
- Flake8: No style violations printed in the final verification.

The following pages contain screenshots captured during the lab session. Some screenshots show the initial findings and others show final verification.

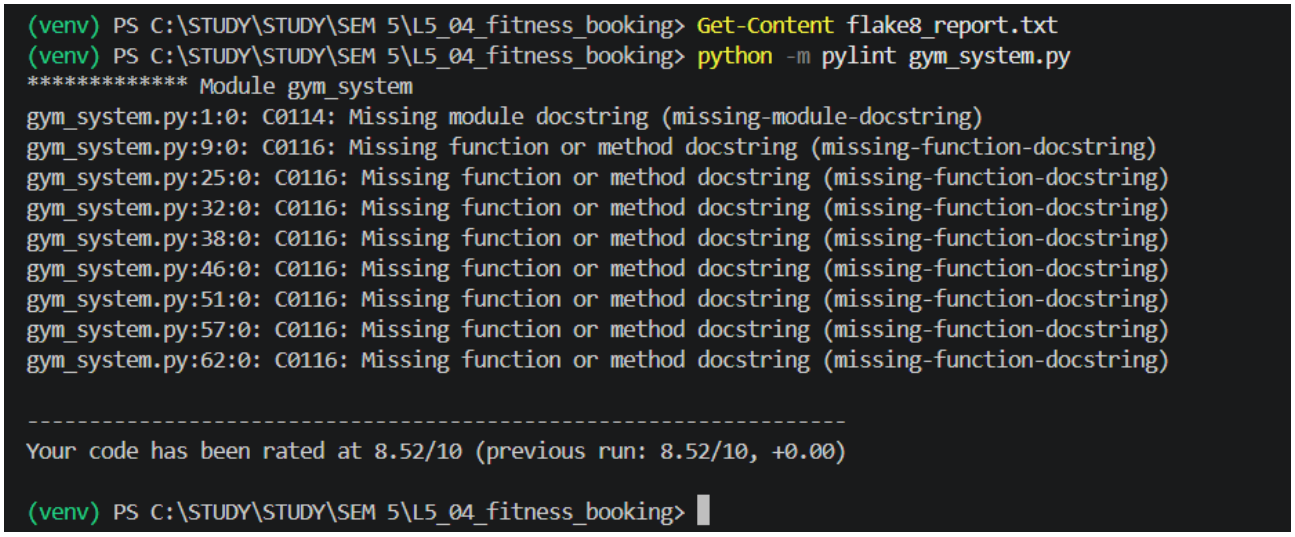
Screenshot 1: Final Pylint score — 10.00/10



Screenshot 2: Bandit final output — no issues identified



Screenshot 3: Pylint and Flake8 verification



Screenshot 4: Final tool reports after fixes

```
(venv) PS C:\STUDY\STUDY\SEM 5\L5_04_fitness_booking> Get-Content bandit_report.txt
python : [main] INFO    profile include tests: None
At line:1 char:1
+ python -m bandit -r gym_system.py > bandit_report.txt 2>&1
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : NotSpecified: ([main]     INFO    profile include tests: None:String) [], RemoteExceptio
n
    + FullyQualifiedErrorId : NativeCommandError

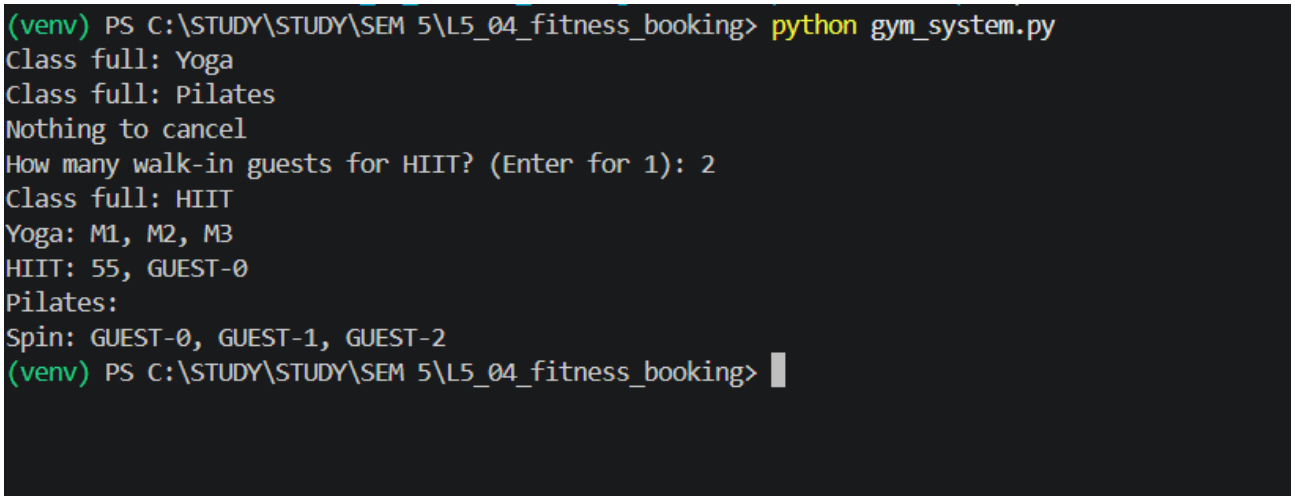
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.8
Run started:2026-10-16 04:06:55.901437+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 61
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Screenshot 5: Corrected program execution



Screenshot 6: Initial static-analysis findings — before fixes

```
(venv) C:\STUDY\STUDY\SEM 5\L5_04_fitness_booking>type pylint_report.txt
************* Module gym_system
gym_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
gym_system.py:8:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:8:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
gym_system.py:15:26: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
gym_system.py:18:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:21:4: W0702: No exception type(s) specified (bare-except)
gym_system.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:27:0: C0103: Function name "addMembers" doesn't conform to snake_case naming style (invalid-name)
gym_system.py:31:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:32:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
gym_system.py:32:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
gym_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:38:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:39:4: C0206: Consider iterating with .items() (consider-using-dict-items)
gym_system.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)

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Your code has been rated at 6.60/10
```

Screenshot 7: Initial static-analysis findings — before fixes

```
(venv) C:\STUDY\STUDY\SEM 5\L5_04_fitness_booking>type bandit_report.txt
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.8
Run started:2026-10-16 04:01:26.532518+00:00

Test results:
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\gym_system.py:36:4
35      def clear_old_exports(prefix="old_"):
36          os.system("rm -f " + prefix + "*.json")
37

--------------------------------------------------

Code scanned:
        Total lines of code: 47
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 1
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 1
Files skipped (0):
```

Screenshot 8: Initial static-analysis findings — before fixes

```
(venv) C:\STUDY\STUDY\SEM 5\L5_04_fitness_booking>type flake8_report.txt
gym_system.py:8:1: E302 expected 2 blank lines, found 1
gym_system.py:18:1: E302 expected 2 blank lines, found 1
gym_system.py:21:5: E722 do not use bare 'except'
gym_system.py:24:1: E302 expected 2 blank lines, found 1
gym_system.py:27:1: E302 expected 2 blank lines, found 1
gym_system.py:31:1: E302 expected 2 blank lines, found 1
gym_system.py:35:1: E302 expected 2 blank lines, found 1
gym_system.py:38:1: E302 expected 2 blank lines, found 1
gym_system.py:42:1: E302 expected 2 blank lines, found 1
gym_system.py:58:1: E305 expected 2 blank lines after class or function definition, found 1

(venv) C:\STUDY\STUDY\SEM 5\L5_04_fitness_booking>more pylint_report.txt
************* Module gym_system
gym_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
gym_system.py:8:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:8:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
gym_system.py:15:26: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
gym_system.py:18:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:21:4: W0702: No exception type(s) specified (bare-except)
gym_system.py:24:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:27:0: C0103: Function name "addMembers" doesn't conform to snake_case naming style (invalid-name)
gym_system.py:31:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:32:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
gym_system.py:32:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
gym_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:38:0: C0116: Missing function or method docstring (missing-function-docstring)
gym_system.py:39:4: C0206: Consider iterating with .items() (consider-using-dict-items)
gym_system.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)

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Your code has been rated at 6.60/10


(venv) C:\STUDY\STUDY\SEM 5\L5_04_fitness_booking>
```

Commands used for verification

```
python -m pylint gym_system.py

python -m bandit -r gym_system.py --format txt

python -m flake8 gym_system.py
```